

VALIDATION REPORT TEMPLATE: SEQUENCING AND BIOINFORMATICS FOR MICROBIAL TARGETS

Title	<u>Viral Genomic Sequencing using Illumina Viral Surveillance Kit – Validation Report</u>
Date	<u>May 12, 2025</u>
Lab Unit	<u>Advanced Molecular Detection Unit</u>
Written by	<u>Jared Johnson, Qing Yang</u>
Proposed Implementation Date	<u>Effective immediately upon approval</u>

APPROVAL:

Upon electronic signature, this validation report will have been reviewed and has been considered acceptable for patient testing. Records should be maintained for the life of the test.

This document describes the validation of a bioinformatic process to analyze whole genome sequencing data to the specified test. Performance characteristics established include, but are not limited to, accuracy and reproducibility for all microbial target types to be detected by the test (for example, organism identification, resistance genes, pathogenicity markers, and host-response markers); the performance characteristics can also include optional metrics.

For each performance characteristic, the following metrics must be included: Accuracy, Precision/Reproducibility, Specificity, Sensitivity. In addition to the required metrics, the following metrics are optional and pertain to population (metagenomic) analyses: PPV, NPV, LOD.

The validation should occur in a same computing environment in which the production results will be generated.

Any usual, unexpected, or significant result must be confirmed with additional testing at the discretion of the laboratory (for example, sending to CDC for confirmation).

Summary of Results:

<u><Each Analytic Result></u>	<u>Expected</u>	<u>Observed</u>	<u>Outcome</u>
<u>Accuracy – Taxonomic Classification</u>	<u>≥ 90% or define otherwise</u>	<u>100%</u>	<u>PASS</u>
<u>Accuracy – Genome Assembly</u>	<u>≥ 90% or define otherwise</u>	<u>95.65%</u>	<u>PASS</u>
<u>Precision/Reproducibility</u>	<u>≥ 95% or define otherwise</u>	<u>100.0%</u>	<u>PASS</u>
<u>Sensitivity (Qualitative Testing)</u>	<u>≥ 95%</u>	<u>Not performed</u>	<u>Not performed</u>
<u>Specificity</u>	<u>≥ 95%</u>	<u>Not performed</u>	<u>Not performed</u>
<u>Optional Metrics (each analytical result)</u>			
<u>Analytical Sensitivity/LOD</u>	<u>Define</u>	<u>Not performed</u>	<u>Not performed</u>

PURPOSE: Please describe **below** the reason for implementing this test and its intended use. Include all acceptable specimen types that will be considered for testing at the PHL. **Note:** Each acceptable specimen type to be tested at the WA PHL must be evaluated. If applicable, attach supporting documents such as package inserts, literature references, and other supporting material used for justifications throughout the validation plan.

Viral genomic surveillance is a cornerstone of infectious diseases surveillance and prevention, providing essential data on the evolution and spread of pathogens. By examining the genetic composition of viruses through next-generation sequencing (NGS), we can detect emerging outbreaks while simultaneously monitoring how viruses evolve and adapt over time. Viral genomic surveillance enables the identification of mutations that can influence critical factors such as transmissibility, virulence, and resistance to antiviral treatments or vaccines. For example, during the COVID-19 pandemic, genomic surveillance played a crucial role in tracking the emergence of new variants such as the Delta and Omicron strains, which exhibited enhanced transmissibility and, in some cases, partial resistance to existing diagnostics, vaccines and treatments. Through continuous monitoring and analysis of these genetic changes, public health authorities can update diagnostic tests, adapt therapeutic strategies, and refine vaccine formulations to stay ahead of evolving threats.

We aim to use the Illumina DNA/RNA Prep with Enrichment with Viral Surveillance Panel v2 kit to detect over 20 virus species that are identified as important risks to public health by the WHO, CDC and WADOH. Once a sample is confirmed positive for the presence of viral genome via previously validated assays that determine the viral taxonomy, it is then submitted to AMD. If the sample is previously extracted using validated methods, AMD will directly perform library preparation and sequencing. Otherwise, the total nucleic acid is first extracted from clinical or environmental samples using Promega Maxwell RSC Viral Total Nucleic Acid Purification kit. The subsequent assay is composed of two major steps: (1) DNA/RNA library preparation followed by sequencing and (2) bioinformatic analysis for genome assembly and taxonomic categorization. The library preparation includes cDNA synthesis step to allow simultaneous detection of both RNA and DNA viral genomes. It uses bead-linked transposome to normalize and fragment diverse input amount from 10 ng to 100ng of total nucleic acid and attach

sequencing adapters. This enables genome sequencing of viral DNA and RNA from diverse sample input types. It then uses biotinylated oligonucleotide probes to hybridize sequencing libraries containing viral sequences. Upon successful enrichment, viral genomes are assembled, and taxonomy is assigned via the WAPHL-developed bioinformatic pipeline, VAPER, using taxon-specific reference genomes developed via the WAPHL-developed EPITOME pipeline.

Selection of Appropriate Method: Please describe how you selected this method, relevant references or standards and how its performance characteristics meet the customer's needs.

For samples that are not previously extracted, we selected Promega Maxwell RSC Viral Total Nucleic Acid Purification kit for input sample extraction. This is an automated method for purifying both viral RNA and DNA from diverse sample input types, including tissue culture isolates, clinical samples in urine, VTM, UTM or PBS and environmental samples. The sample purification is performed using Promega CSC 48 instrument with minimum hands-on procedures, enabling simultaneous purification of 48 samples within 30 minutes. This method has been previously validated for extraction Mumps virus nucleic acid from samples in VTM and urine (69790.5045 Mumps Virus Detection by Real-time RT-PCR SOP). Together, because of its compatibility with diverse sample types, high-throughput automation and existing validation, we determined the Promega Maxwell purification kit to be the optimal method for sample extraction.

Next, the Illumina DNA/RNA Prep with Enrichment using Viral Surveillance Panel v2 kit uses well-established methods to generate Illumina sequencing libraries. The key library preparation step has been validated for multiple whole genome sequencing applications here at WAPHL and CDC (MediaLab #69790.3690, #69790.2280). The library preparation steps are multiplex- and automation-compatible with minimal hands-on time. Compared to other enrichment methods (e.g. amplicon sequencing), the probe-hybridization-based enrichment for viral sequences is advantageous in ensuring a uniform coverage across viral genome and in capturing mutations (Sahajpal *et al.* 2021, Ceballos-Garzon *et al.*, 2024). The enrichment probes target more than 200 viruses that encompass all viruses of interest at WAPHL. Together, this assay was selected because of the robustness of established library preparation techniques, the hybrid-capture-based enrichment and the comprehensiveness of the viral targets.

VAPER employs the same general, reference-based assembly method as the WAPHL-validated genome assembly pipeline for SARS-CoV-2 from tiled-amplicon data (MediaLab Document # 69790.2002). The key distinction between these pipelines is how reference genomes are selected. The SARS-CoV-2 pipeline is designed for a single viral species (i.e., Severe acute respiratory syndrome coronavirus 2), therefore requiring a single, species-specific reference genome. In contrast, VAPER is designed to handle multiple viral species, therefore requiring multiple species-specific references. Reference selection is one of the most crucial determinants of genome assembly quality (Valiente-Mullor *et al.*, 2021). For this reason, the auxiliary pipeline, EPITOME, was created to facilitate the creation of high-quality reference genomes that capture the known diversity of a viral species at discrete divergence levels (e.g., 5% average nucleotide difference). Together, EPITOME and VAPER provide a framework for assembling any viral species captured by probe-based enrichment.

TECHNIQUE: Please describe **below** the scientific basis of how this test works.

Overview

The [viral genomic surveillance assay using Illumina Viral Surveillance Panel V2 kit](#) includes an optional sample extraction step, followed by sequencing and bioinformatic analysis of nucleic acid containing viral genomes. The sequencing step includes cDNA synthesis, DNA library preparation and library enrichment via probe-based hybrid capture. The enriched library can be sequenced on multiple Illumina MiSeq platforms currently available at the WAPHL. Upon sequencing completion, the raw sequencing files in FASTQ format are analyzed through the bioinformatic workflow, VAPER, developed at the WAPHL. This workflow will provide sequencing QC data, taxonomic categorization and genome assembly of viruses.

Total Nucleic Acid Extraction with Promega Maxwell RSC Viral Total Nucleic Acid Purification kit

For unextracted samples stored in VTM/UTM or other storage media, AMD will perform total nucleic acid extraction using the Promega Maxwell system. The Promega Maxwell RSC Viral Total Nucleic Acid Purification kit relies on paramagnetic bead-based purification. The input sample is first mixed with lysis buffer containing mild detergent, proteinase K and guanidine hydrochloride to release and stabilize nucleic acid content, while degrading other lipid and protein components of the virus. The lysate is then added to a cartridge and loaded onto Promega liquid handler. The cartridge contains paramagnetic purification beads that attract nucleic acid via the carboxyl molecule coating. During this step, the binding buffer efficiently adjusts the pH and provides salt ions to allow efficient nucleic acid binding, regardless of the input sample types (tissue culture, mosquito lysates, urine, UTM, VTM or PBS). The bead-bound nucleic acid is then cleaned-up by binding to a plunger with a magnetic rod. After a series of washes, purified total nucleic acid is eluted into storage buffer for downstream sequencing.

Library preparation with Illumina DNA/RNA Prep with Enrichment using Viral Surveillance Panel v2

To process the extracted total nucleic acid into sequencing library using the Illumina DNA/RNA Prep kit, RNA content is first denatured and reverse transcribed into cDNA using random hexamer primers. The single-stranded cDNA is then converted to double-stranded DNA using the second strand synthesis master mix (**Figure 1A**). The cDNA and the pre-existing DNA are tagmented using Illumina bead-linked transposome, which fragments DNA into uniform sizes and attaches sequencing primers to both ends of the DNA. The tagmented DNA library is then amplified using index primers through a PCR reaction. This step adds unique dual indexes to each of the samples. After PCR amplification, the library is cleaned up using paramagnetic beads to remove unincorporated nucleotides, primers and small amplicons. Before hybridization, the sequencing libraries are quantified using Qubit reagent on SpectraMax plate reader, the libraries that pass quality check will be included in the downstream processes.

To enrich the virus-specific sequencing library, the libraries are processed using probe-based hybridization (**Figure 1B**). The libraries are first pooled into groups of three to allow multiplex in the downstream steps. The sequencing libraries are denatured, hybridized to biotinylated oligonucleotide probes targeting over 200 different viral genomes, and attracted to streptavidin-coated magnetic beads. This hybridization followed by wash steps allows enrichment of sequencing libraries containing only viral genome sequences. The enriched sequencing library is further amplified via PCR reaction to ensure sufficient coverage of viral genomes. The final sequencing library undergoes another bead-based clean up to remove excess primers, nucleotides and primer dimers. The final

concentration of the enriched sequencing library is quantified using DNA Qubit reagent on SpectroMax plate reader. The libraries that pass the quality check are pooled and analyzed on the TapeStation to determine the average DNA fragment size. Based on the fragment size and concentration, the final pooled library is diluted and loaded onto an Illumina MiSeq sequencing platform.

Illumina MiSeq sequencing instrument uses sequencing-by-synthesis technology where fluorescently labeled nucleotides are sequentially hybridized to the library to allow base identification. This platform can provide sufficient coverage for diverse sample types and sizes. Using different reagent cartridges and flow cells, MiSeq is adequate for providing 1 x 36 to 2 x 300 base pair read length and 1.2 – 14 Gb output. For routine sequencing of viral genomes, 2 x 150 read length configuration is most suitable to achieve desired coverage and quality criteria. For more information on The Illumina MiSeq platforms see MediaLab document # 69790.3163.

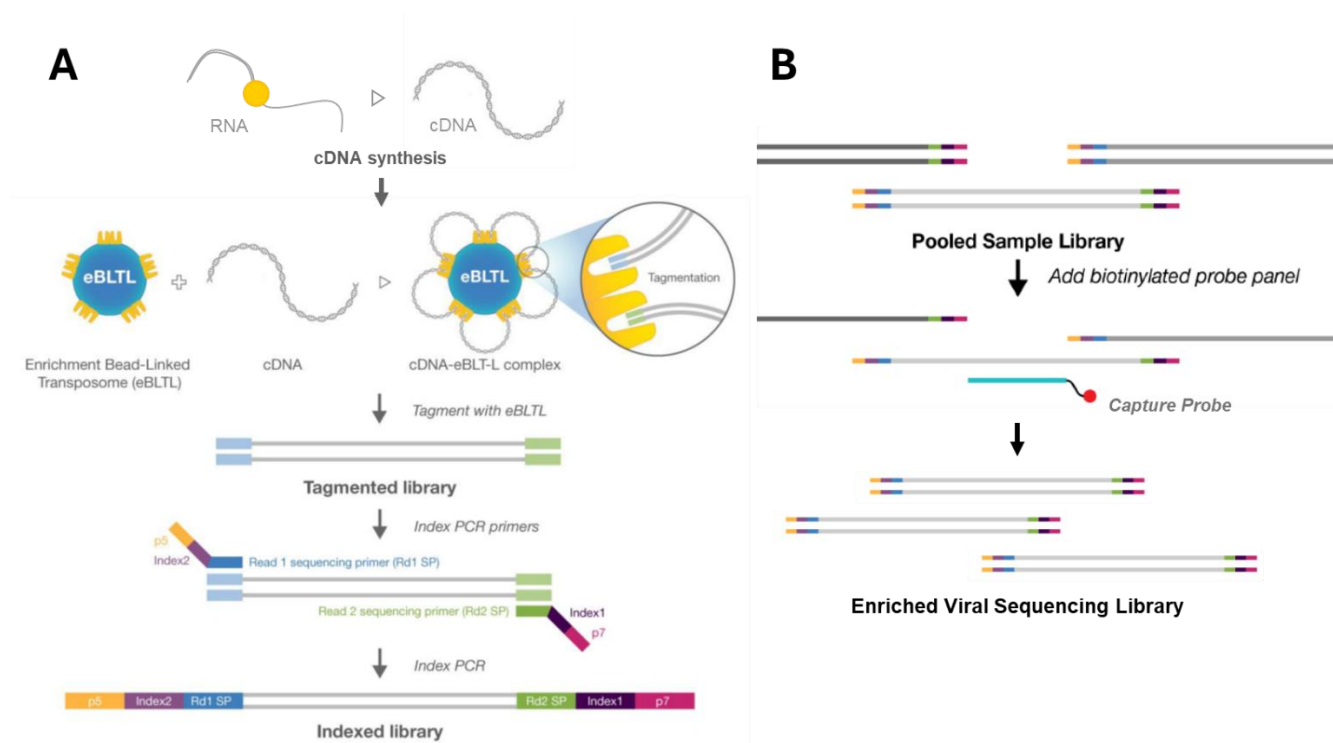


Figure 1. An overview of the library preparation process of Illumina DNA/RNA Prep with Enrichment using Viral Surveillance Panel v2. (A) Input extracted RNA is first reverse transcribed into cDNA. The cDNA and extracted dsDNA are bound to enrichment bead-linked transposome, which fragments and attaches sequencing adapters to DNA. The individual library is further amplified and indexed through PCR. **(B)** Upon library generation, the libraries are pooled into triplex and incubated with biotinylated probes to capture virus sequences. This results in enriched viral sequencing library with minimum host or environmental nucleic acid contamination.

Reference-based genome assembly by VAPER

Viral genomes will be assembled using the same general, reference-based approach used in the validated method for SARS-CoV-2 genome assembly (MediaLab Document # 69790.2002). High quality reads are aligned to a reference genome and the nucleotide identity most supported by the reads is reported at each reference position.

In the case that a nucleotide identity cannot be confidently called, an ambiguous nucleotide code is returned (i.e., R, K, S, W, Y, M, B, H, D, V, or N).

Reference selection can greatly impact the quality of reference-based genome assemblies. Work conducted at WA PHL demonstrated that a decrease in genome assembly quality is observed when the reference differs from the sample by roughly 5% or greater average nucleotide identity (Figure 2). Given that many viruses exhibit intra-species divergence > 5% (Peterson *et al.*, 2014), it was anticipated that multiple reference genomes would be required per species to ensure consistent genome assembly quality. This challenge was addressed via the development of the auxiliary pipeline, EPITOME (<https://github.com/DOH-JDJ0303/EPITOME>), which creates compressive reference sets from a collection of input sequences at a divergence threshold. Reference genomes were created from 637,636 complete genome sequences, sourced from NCBI and GISAID, belonging to 529 viral species and 18 viral genera: *Alphacoronavirus*, *Alphainfluenzavirus*, *Betacoronavirus*, *Betainfluenzavirus*, *Bocaparvovirus*, *Enterovirus*, *Hantavirus*, *Hepacivirus*, *Hepatovirus*, *Lyssavirus*, *Mastadenovirus*, *Metapneumovirus*, *Morbillivirus*, *Orthoflavivirus*, *Orthopneumovirus*, *Orthopoxvirus*, *Orthorubulavirus*, and *Respirovirus*. Reference sets were initially created using a 5% divergence threshold; however, this threshold was found to be too liberal for taxonomic classification, resulting in approximately 4% of the references being associated with two or more species classifications. As such, a more conservative 2% threshold was used, reducing the relative number of multi-species references by approximately 50% (~4% to ~2%). VAPER automatically selects reference genomes that most closely match the sample by comparing contigs generated via de novo assembly using competitive alignment (i.e., only one, best match per contig). This approach allows for the possibility to select multiple reference genomes per sample, thus making it possible to assemble samples containing segmented viruses and/or multiple viral species differing by at least 2% average nucleotide identity.

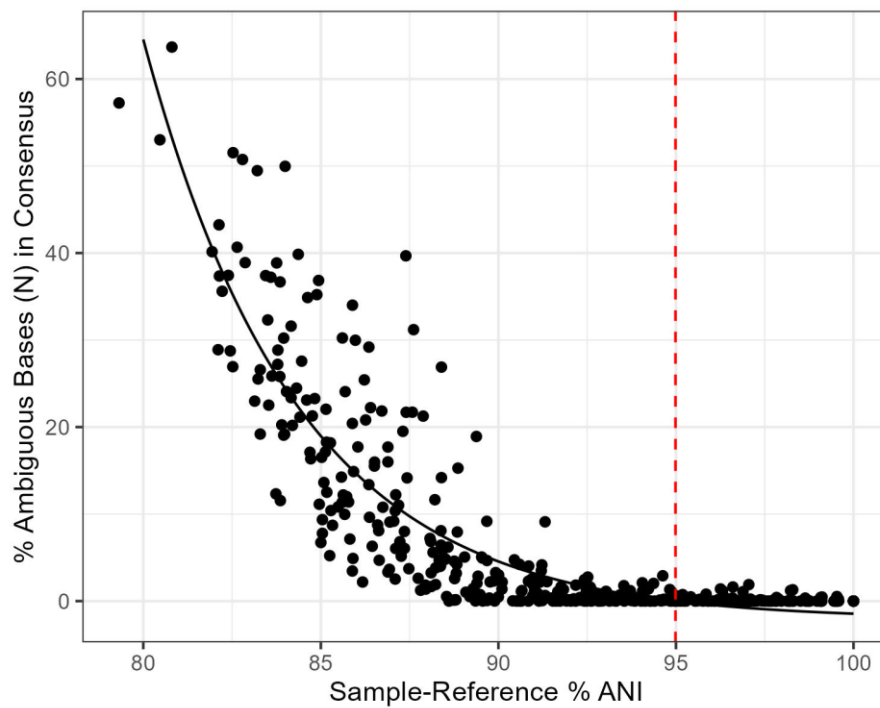


Figure 2. Relationship between sample-to-reference divergence and genome assembly quality when creating reference-based viral genome assemblies. Genome assemblies were created using references contrived from 5 sequences (OK573275.1, MF965288.1, OQ096316.1, PP175946.1, PP175947.1), representing 4 viral species (West Nile, Mumps, Measles, and Influenza A). References were contrived, in duplicate, by introducing mutations at a rate of 0 to 21% nucleotide divergence, with a step increase of 0.5% divergence, using Varcraft (<https://github.com/DOH-HNH0303/Varcraft/>). Regression analysis determined ambiguous base rates of approximately ~65, ~19, ~5, and ~0% in the final assembly when using references that diverged by 20, 15, 10, and 5% nucleotide identity. The red, dashed line represents the 5% divergence threshold estimated to minimize loss in assembly quality.

Taxonomic Classification by VAPER

VAPER derives the taxonomy of each genome assembly using the classification associated with each reference genome. Reference taxonomy is based on the taxonomy of individual sequences used to create the reference, as reported to NCBI or GISAID. Given that viral taxonomy is not standardized (Peterson *et al.*, 2014), it is possible for a single species to be represented by multiple references and for a single reference to represent multiple species. As such, it is possible that a single assembly will have multiple associated taxonomic classifications.

INSTRUMENTATION & REAGENTS: Please describe **below** the reagents/ kit and instrumentation that will be used in the validation. Add rows below as needed.

Illumina Viral Surveillance Panel V2 Components	Set A		Set B	
	Lot Number	Expiration Date	Lot Number	Expiration Date
ILMN DNA/RNA Prep, Tag Buffers 96	20926120	1/5/2026	20931700	1/11/2026
ILMN DNA/RNA Prep, Tag Beads 96	20926582	1/16/2026	20931557	1/16/2026
ILMN DNA/RNA Prep, Tag PCR Reagents 96	20921140	12/24/2025	20926553	1/2/2026
ILMN DNA/RNA Fast Hyb Enrichm Beads 96	20933581	1/17/2026	20931554	1/17/2026
ILMN RNA Sing Hyb Enrich PCR 96	20918565	11/30/2025	20918565	11/30/2025
ILMN cDNA Sythesis 96	20930242	11/22/2025	20941046	1/6/2026
Illumina Purification Beads 3x10mL	20920727	12/10/2026	20920727	12/10/2026
Viral Surveillance Panel V2	20928242	12/18/2026	20928242	12/18/2026
ILMN DNA/RNA UD Index Set A	20933858	1/13/2027	20933859	1/14/2027

Additional Reagents	Lot Number	Expiration Date
Promega Maxwell Viral TNA Purification Kit	14597	2/1/2027
Agilent High Sensitivity D5000 ScreenTape and Reagent	0203636; 0006825561	3/10/2026; 12/9/2025
Qubit DNA High Sensitivity Assay Kit	2856039	6/5/2025
Molecular Water	23H0756009	8/14/2027
PhiX Control	20875428	6/7/2025
NaOH	012225AD	08/2026
MiSeq Cartridge	20909307	9/22/2025
	20912643	9/29/2025

MiSeq Flow cell	20885479	07/27/2025
	20931226	12/24/2025
	20923469	12/24/2025
	20921918	12/17/2025
	20891298	8/26/2025
	20931210	1/27/2026

Instrument	Serial Number	Manufacturer
MiSeq M05130	M05130	Illumina
MiSeq M05916	M05916	Illumina
MiSeq M08782	M08782	Illumina
Agilent 4200 TapeStation System	DEDAA03608	Agilent Technologies
SpectraMax M2 Plate Reader	D05494	Molecular Devices
Promega Maxwell CSC 48 System	21000160	Promega

COMPUTING ENVIRONMENT/SOFTWARE PACKAGES: Please describe **below** the computing environment, software package(s), script(s), and/or database(s) that will be used in the validation. Note: 69790.3152 Bioinformatics – IT Infrastructure Manual should contain most of the discussion on the computing environment; refer to appropriate section within the manual.

Sequence analysis was conducted using Amazon Web Services (AWS) elastic container (EC2) services, as described in the 69790.3152 Bioinformatics – IT Infrastructure Manual. Software was executed using containerized images described in the table below, thus guaranteeing reproducibility between runs. Genome assembly and taxonomic assignment will be accomplished using the “EPITOME_2025-02-06 reference set (https://github.com/DOH-JDJ0303/vaper/blob/main/assets/reference_sets/EPITOME_2025-02-06.tar.gz)”.

Pipeline Process	Container
SAMPLESHEET_CHECK	biocontainers/python:3.8.3
TAR2REFS, SUMMARIZE_TAXA, SUMMARYLINE, COMBINE_SUMMARYLINES	public.ecr.aws/o8h2f0o1/vaper-base:1.0
SEQTK_SAMPLE, MAPPED_FASTQ	biocontainers/seqtk:1.3--h5bf99c6_3
FASTQC	biocontainers/fastqc:0.11.9--0
FASTP	biocontainers/fastp:0.23.4--h5f740d0_0
CONDENSE	biocontainers/sourmash:4.8.4--hdfd78af_0
FORMAT_REFS	staphb/seqtk:1.3
SHOVILL	biocontainers/shovill:1.1.0--0
MINIMAP2_ALIGN	biocontainers/mulled-v2-66534bcb7031a148b13e2ad42583020b9cd25c4:1679e915ddb9d6b4abda91880c4b48857d471bd8-0
BWA_MEM, BAM_STATS	biocontainers/mulled-v2-fe8faa35dbf6dc65a0f7f5d4ea12e31a79f73e40:219b6c272b25e7e642ae3ff0bf0c5c81a5135ab4-0
IVAR_CONSENSUS	public.ecr.aws/o8h2f0o1/ivar:1.4.2
NEXTCLADE	docker.io/nextstrain/nextclade:3.9.1

SAMPLES: Describe the breadth of samples for the validation including the following criteria (additional criteria are acceptable depending on the parameters of the assay and as approved by the WA PHL CLIA Director):

The samples for this validation consist of 80 clinical samples sourced from University of Washington and Virology at WAPHL, 10 environmental samples from ELS at WAPHL, and 17 synthetic control samples from Twist Biosciences or BEI resources (**Figure 3**). Please see **Appendix A** for detailed description of the sample and virus taxonomy. Please refer to section “REQUIRED ANALYSIS METRICS” for detailed description on the usage of the samples.

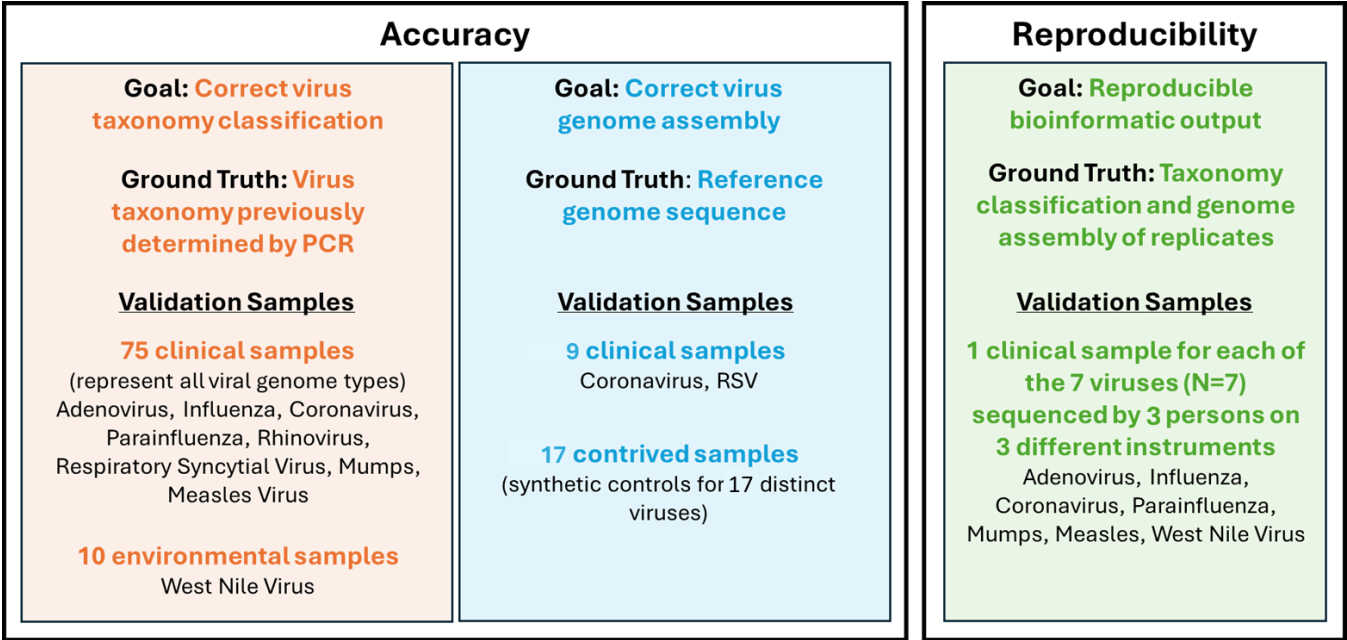


Figure 3. Summary of the samples to be used for validation. A combination of clinical, environmental and contrived samples was used for validation. A large set of samples with taxonomic-specific PCR results was used to evaluate the assay’s ability to correctly categorize the virus taxonomy. To further evaluate the assay’s ability to generate genome assembly correctly, additional clinical samples and contrived samples with published references sequences were used for comparison. Finally, for reproducibility evaluation, a subset of the clinical samples was sequenced by three trained personnels on three separate days, and the bioinformatic output was compared for correct classification and genome assembly.

SEQUENCE DATABASE DESCRIPTION

Describe the sequence database(s) containing the sequences and identifiers of the target organisms, resistance and pathogenicity markers, and host-responsive markers. Depending on the application, the database description may include representation of taxonomic groups, rare species, potentially confounding identical genotypes in clinically dissimilar organisms, potentially contaminating environmental and microbiota organisms, and the degree of taxonomic curation of the database. Resistance databases must be documented as to drugs and drug classes represented, and pathogen strains and/or genotypes, as clinically indicated.

Sequence Database	Description
Database name	EPITOME_2025-02-06
Taxonomic groups	6396 reference genomes representing 519 species belonging to 18 genera (<i>Alphacoronavirus</i> , <i>Alphainfluenzavirus</i> , <i>Betacoronavirus</i> , <i>Betainfluenzavirus</i> , <i>Bocaparvovirus</i> , <i>Enterovirus</i> , <i>Hantavirus</i> , <i>Hepacivirus</i> , <i>Hepatovirus</i> , <i>Lyssavirus</i> , <i>Mastadenovirus</i> , <i>Metapneumovirus</i> , <i>Morbillivirus</i> , <i>Orthoflavivirus</i> , <i>Orthopneumovirus</i> , <i>Orthopoxvirus</i> , <i>Orthorubulavirus</i> , <i>Respirovirus</i>)
Degree of curation	Reference genomes were created from 579,102 sequences, sourced from NCBI and GISAID, using the EPITOME v1.5 pipeline with a 0.02 average nucleotide divergence threshold. Sequences were included if they met the following criteria: - Listed as “complete” in GISAID or NCBI < 2% ambiguous characters (i.e., “N”) < 20% deviation from the median taxon-segment sequence length - Had a specified segment name (segmented viruses only)

QUALITY METRICS

Wet-bench Method Metrics Acceptance and Rejection Criteria:

These should include criteria for determining when the sequencing process has failed and whether the data should be either re-processed or not further processed.

Pre-Run Acceptance/Rejection Criteria

- Negative control library at the pre-hybridization step with a DNA concentration < 1.0 ng/μL, determined via SpectraMax using Qubit High Sensitivity DNA reagent.
- Final post-hybridization pooled library concentration ≥ 1.0 ng/μL, determined via SpectraMax using Qubit High Sensitivity DNA reagent.
- Final post-hybridization pooled library average fragment size ≥ 250 bp, determined via Tapestation using the High Sensitivity D5000 reagent.

Post-Run Acceptance/Rejection Criteria

Illumina MiSeq Metrics	
Sequencing Chemistry	300c V2
Cluster Density	800 – 1400 k/mm^2
Recommended %Q30 (per Sample)	≥ 80%
Cluster passing filter	>80%
Estimated total yield (Gbp)	> 5.5

Dry bench/ Post assembly Metrics Acceptance and Rejection Criteria:

Criteria must determine when the post-assembly data is sufficiently inadequate and requires the sample to be either re-processed or not further processed.

Post-Assembly Acceptance/Rejection Criteria

Criteria will be evaluated relative to the reference genome for each sample assembly.

- ≥ 30X depth of coverage

- $\geq 80\%$ genome recovery
- No assembly generated for the negative control

ANALYSIS PROCESS:

Please describe **below** the sequential process that will be used in the analysis of the sequence reads; include list of scripts and when databases references will be utilized, focus of analytical process, and output. Is the analysis on a population (for example, a metagenomic analysis)? Will the results of the process be for surveillance efforts? Will the results be reported back to specimen submitter? Note: this section will also include any confirmatory testing that was conducted during the validation process. Examples of independent methods for confirmation include Sanger sequencing, pyrosequencing, targeted nucleic acid amplification testing, and culture and phenotypic antimicrobial susceptibility testing results.

Viral Assembly from Probe-based Enrichment (VAPER)

Viral assemblies are created from probe-enriched reads using a reference-based approach. First, raw reads are quality filtered, and sequencing adapters removed using Fastp (Chen et al., 2018). High-quality reads are then de novo assembled using MegaHit (Li et al., 2015) via Shovill (<https://github.com/tseemann/shovill>) and the contigs are competitively aligned (one alignment per contig) to the EPITOME_2025-02-06 reference genomes using Minimap2 (Li et al., 2018), allowing for up to approximately 5% sample-reference nucleotide divergence. References with at least 10% genome coverage by one or more sample contig are selected by VAPER for reference-based consensus assembly. High-quality reads are then aligned to each reference genome, separately, using BWA MEM (Li et al., 2009) and a consensus sequence is generated using Samtools (Danecek et al., 2021) and iVar (Grubaugh et al., 2019). Assembly quality is evaluated using Nextclade (Aksamentov et al., 2021) and custom scripts written in R. Assembly taxonomy is derived from the metadata associated with each EPITOME reference genome. Sample metrics, including read quality, assembly quality, and the predicted taxonomy of each viral assembly are summarized using custom R scripts and saved to a tab-separated file.

VAPER v1.0.0

<https://github.com/DOH-JDJ0303/vaper>

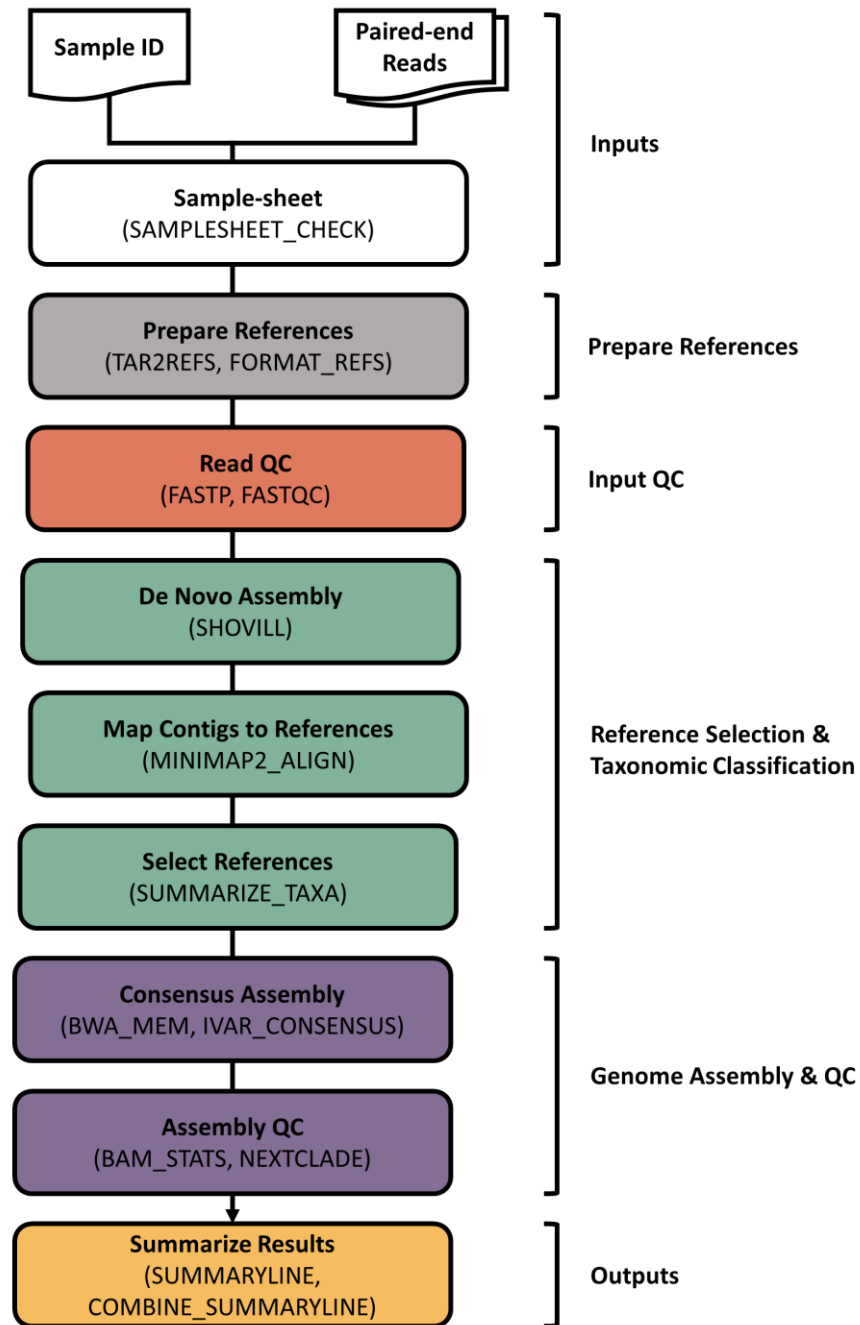


Figure 4. An overview of the VAPER pipeline.

ANALYTICAL SPECIFICITY/INTERFERENCE: Analytical Specificity refers to the ability of a test or procedure to correctly identify or quantify an entity in the presence of interfering or cross-reactive substances that might be expected to be present.

Site any literature references of known interferences and/or perform standard addition type experiments to determine the factors that may pose significant issues for the assay specificity.

Please describe **below** your plan to establish the analytical specificity of the assay.

Each virus species that is not the input target organism will be considered a potential cross-reacting species. The evaluation is based on the ability of the system to correctly identify the input.

Sequence characteristic interference: How will interfering characteristics, such as highly homologous regions and low complexity regions, be mitigated?

Interfering characteristics, such as highly homologous regions, would be anticipated to result in two adverse outcomes:

- The first outcome would be the misidentification of nucleotides in VAPER assemblies resulting from the misalignment of reads to homologous regions of the reference genome. This is mitigated by the strict quality criteria employed during the consensus assembly process, which requires that each nucleotide be supported by at least 10 reads with a Phred quality score of 20 or greater. Further, at least 50% of the reads must support a single nucleotide identity or an ambiguous nucleotide code (i.e., R, K, S, W, Y, M, B, H, D, V, or N) is returned. Nucleotide positions supported by fewer than 10 reads meeting this criterion would be reported as unknown, as represented by nucleotide code N.
- The second outcome would be the misclassification of assembly taxonomy resulting from the misalignment of homologous contigs during the automated reference selection process. This is mitigated by the strict divergence threshold employed during the reference creation process, which requires that each reference is created from sequences that differ by no more than 2% of their shared (core) regions, therefore resulting in contrived references that differ by 2% or more of their shared regions. In the case that a reference was created from sequences belonging to two different viral species, both species classifications would be returned.

REQUIRED ANALYSIS METRICS

Clearly define the standard(s) to be used to measure accuracy. If the standard is a reference laboratory, that the reference lab is CLIA certified and validated for the assay. Clearly define the measurements to determine accuracy and reproducibility (for example cg MLST, numbers of SNPs, etc.).

ACCURACY: Are your test results correct? This can be determined by testing previously annotated genomes, side-by-side testing with a reference laboratory, proficiency testing materials that have established values, and previously tested patient specimens with established values. Total accuracy should be **at least 90%** relative to the reference method.

$$\text{Accuracy (Agreement)} = \frac{\text{No. of correct results}}{\text{Total no. of results}} \times 100$$

$$\text{Taxonomy Classification Accuracy} = 100 \times \frac{168}{168} = 100\%$$

$$\text{Genome Assembly Accuracy} = 100 \times \frac{44}{46} = 95.7\%$$

The overarching goal for this validation is to show that the Promega Maxwell total nucleic acid extraction, Illumina Viral Surveillance kit along with the custom bioinformatic pipeline VAPER can reliably classify the virus taxonomy and assemble the viral genome from diverse input samples. However, due to the limited methods available for viral genome sequencing, most of the clinical samples were confirmed via ddPCR or qRT-PCR methods but lack sequencing data. To overcome this challenge, we validated this assay using a combination of clinical, environmental and contrived samples (**Figure 3 and Appendix A**).

Taxonomic Classification

To ensure that the assay can correctly categorize the virus taxonomy from input samples, we used the taxonomic information previously determined by RT-PCR/PCR assays as the ground truth. Provided by the manufacturer, the taxonomic information is elucidated from the primer and probe sequence of the RT-PCR assays, which differentiates virus family, species and/or strains (**Table 1**). We validated 85 infection samples, including 75 clinical samples (with Adenovirus, Coronavirus, Influenza virus, Parainfluenza virus, Rhinovirus, Respiratory Syncytial Virus, Mumps virus, or Measles virus infections) and 10 environmental samples (containing West Nile virus).

In this comparison, we first determined whether the sample has sufficient sequencing data to generate viral genome assembly that met QC criteria (indicated in the status column as “Accept” or “Reject”). For accepted samples, we then used the complete genome assembly to classify virus taxonomy. The accuracy of taxonomic classification by VAPER was evaluated as the percentage of classifications matching the expected result for each sample shown in **Table 1**. In the case of segmented virus genomes (e.g. Influenza virus), individual genome assembly for each segment was generated and compared for taxonomy classification separately. In total, 185 viral genomes/segments were expected, and 168 viral genome segments were successfully assembled (Accepted). Among the 17 rejected viral genomes that failed to meet assembly QC criteria, 16 genomes were originally indicated as low-abundance co-infecting viruses in the clinical samples (Sample #76- #90) with Crt values failed to meet the sample acceptance criteria. Regardless, we were able to generate genome assemblies for the dominant viruses within Sample #76- #90, with correct taxonomic comparisons results. Moreover, one environmental sample (Sample #70) containing West Nile Virus failed assembly QC and was excluded from the comparison. Among all 168 accepted viral genomes assemblies, the taxonomic classification demonstrated 100% accuracy.

Table 1: Virus Classification Accuracy reporting table

Sample #	Viral Genome Form	qRT-PCR/ddPCR Target	Ct/Crt* values	Accepted Taxonomic Result**	Observed Taxonomic Result	Analysis QC	Interpretation
1	dsDNA	AdV_1of2 Crt 1	3.75 / 3.52	Human mastadenovirus B; Human mastadenovirus C; Human mastadenovirus D	[Human mastadenovirus B; Human adenovirus sp.]	Accept	Correct
2	dsDNA	AdV_1of2 Crt 1	3.74 / 3.7	Human mastadenovirus B; Human mastadenovirus C; Human mastadenovirus D	[Human mastadenovirus B; Human adenovirus sp.]	Accept	Correct
3	dsDNA	AdV_1of2 Crt 1	4.79 / 4.69	Human mastadenovirus B; Human mastadenovirus C; Human mastadenovirus D	[Human mastadenovirus B; Human adenovirus sp.]	Accept	Correct

4	dsDNA	AdV_1of2 Crt 1	4.33/ 4.25	Human mastadenovirus B; Human mastadenovirus C; Human mastadenovirus D	[Human mastadenovirus B; Human adenovirus sp.]		Accept	Correct
5	dsDNA	AdV_1of2 Crt 1	4.78 / 4.83	Human mastadenovirus B; Human mastadenovirus C; Human mastadenovirus D	[Human mastadenovirus C; Human adenovirus 108; Human adenovirus sp.]		Accept	Correct
6	dsDNA	AdV_2of2 Crt 1	3.83/ 3.81	Human mastadenovirus B; Human mastadenovirus C; Human mastadenovirus D	[Human mastadenovirus B; Human adenovirus sp.]		Accept	Correct
7	dsDNA	AdV_2of2 Crt 1	4.51/ 4.33	Human mastadenovirus B; Human mastadenovirus C; Human mastadenovirus D	[Human mastadenovirus B; Human adenovirus sp.]		Accept	Correct
8	dsDNA	AdV_2of2 Crt 1	4.86/ 4.75	Human mastadenovirus B; Human mastadenovirus C; Human mastadenovirus D	[Human mastadenovirus B; Human adenovirus sp.]		Accept	Correct
9	dsDNA	AdV_2of2 Crt 1	10.83 / 10.76	Human mastadenovirus B; Human mastadenovirus F; Human mastadenovirus A; unidentified adenovirus; Human mastadenovirus E; Human adenovirus sp.	[Human mastadenovirus E; Human adenovirus sp.]		Accept	Correct
10	dsDNA	AdV_2of2 Crt 1	12.27 / 12.19	Human mastadenovirus B; Human mastadenovirus F; Human mastadenovirus A; unidentified adenovirus; Human mastadenovirus E; Human adenovirus sp.	[Human mastadenovirus E; Human adenovirus sp.]		Accept	Correct
11	(+)ssRNA	CoV_229E Crt 1	5.71 / 5.84	Human coronavirus 229E	Human coronavirus 229E		Accept	Correct
12	(+)ssRNA	CoV_229E Crt 1	6.01 / 5.9	Human coronavirus 229E	Human coronavirus 229E		Accept	Correct
13	(+)ssRNA	CoV_HKU1 Crt 1	8 / 7.96	Human coronavirus HKU1	Human coronavirus HKU1		Accept	Correct
14	(+)ssRNA	CoV_HKU1 Crt 1	11.86 / 11.85	Human coronavirus HKU1	Human coronavirus HKU1		Accept	Correct
15	(+)ssRNA	CoV_NL63 Crt 1	9.82 / 9.74	Human coronavirus NL63	Human coronavirus NL63		Accept	Correct
16	(+)ssRNA	CoV_NL63 Crt 1	8.43 / 8.48	Human coronavirus NL63	Human coronavirus NL63		Accept	Correct
17	(+)ssRNA	CoV_OC43 Crt 1	5.81 / 5.72	Betacoronavirus 1	Betacoronavirus 1		Accept	Correct
18	(+)ssRNA	CoV_OC43 Crt 1	6.47 / 6.56	Betacoronavirus 1	Betacoronavirus 1		Accept	Correct
19	(+)ssRNA	SARS-CoV-2_Orf1b Crt 1	6.04 / 6.13	Severe acute respiratory syndrome coronavirus 2	Severe acute respiratory syndrome coronavirus 2		Accept	Correct
20	(+)ssRNA	SARS-CoV-2_Orf1b Crt 1	6.12 / 6.09	Severe acute respiratory syndrome coronavirus 2	Severe acute respiratory syndrome coronavirus 2		Accept	Correct
21	Segmented (-)ssRNA	Flu_A_H1 Crt 1	7.37 / 7.47	Alphainfluenzavirus influenzae	Segment 1	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 2	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 3	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 4	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 5	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 6	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 7	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 8	Alphainfluenzavirus influenzae	Accept	Correct
22	Segmented (-)ssRNA	Flu_A_H1 Crt 1	12.48 / 12.5	Alphainfluenzavirus influenzae	Segment 1	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 2	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 3	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 4	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 5	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 6	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 7	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 8	Alphainfluenzavirus influenzae	Accept	Correct
23	Segmented (-)ssRNA	Flu_A_H1 Crt 1	12.82 / 12.84	Alphainfluenzavirus influenzae	Segment 1	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 2	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 3	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 4	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 5	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 6	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 7	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 8	Alphainfluenzavirus influenzae	Accept	Correct
24	Segmented (-)ssRNA	Flu_A_H3 Crt 1	8.38 / 8.24	Alphainfluenzavirus influenzae	Segment 1	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 2	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 3	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 4	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 5	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 6	Alphainfluenzavirus influenzae	Accept	Correct

[illegible]

				Orthorubulavirus laryngotracheitidis				
36	(-)ssRNA	hPIV3_hPIV4 Crt 1	3.37 / 3.66	Respirovirus pneumoniae; Orthorubulavirus hominis	Respirovirus pneumoniae		Accept	Correct
37	(-)ssRNA	hPIV3_hPIV4 Crt 1	6.35 / 6.27	Respirovirus pneumoniae; Orthorubulavirus hominis	Respirovirus pneumoniae		Accept	Correct
38	(-)ssRNA	hPIV3_hPIV4 Crt 1	6.78 / 6.71	Respirovirus pneumoniae; Orthorubulavirus hominis	Respirovirus pneumoniae		Accept	Correct
39	(-)ssRNA	hPIV3_hPIV4 Crt 1	6.62 / 6.61	Respirovirus pneumoniae; Orthorubulavirus hominis	Respirovirus pneumoniae		Accept	Correct
40	(-)ssRNA	hPIV3_hPIV4 Crt 1	6.66 / 6.69	Respirovirus pneumoniae; Orthorubulavirus hominis	Respirovirus pneumoniae		Accept	Correct
41	(-)ssRNA	Mumps RT-PCR	25.24	Orthorubulavirus parotidis	Orthorubulavirus parotitidis		Accept	Correct
42	(-)ssRNA	Mumps RT-PCR	25.64	Orthorubulavirus parotidis	Orthorubulavirus parotitidis		Accept	Correct
43	(-)ssRNA	Mumps RT-PCR	25.81	Orthorubulavirus parotidis	Orthorubulavirus parotitidis		Accept	Correct
44	(-)ssRNA	Mumps RT-PCR	25.74	Orthorubulavirus parotidis	Orthorubulavirus parotitidis		Accept	Correct
45	(-)ssRNA	Mumps RT-PCR	23.88	Orthorubulavirus parotidis	Orthorubulavirus parotitidis		Accept	Correct
46	(-)ssRNA	Mumps RT-PCR	23.86	Orthorubulavirus parotidis	Orthorubulavirus parotitidis		Accept	Correct
47	(-)ssRNA	Mumps RT-PCR	24.86	Orthorubulavirus parotidis	Orthorubulavirus parotitidis		Accept	Correct
48	(-)ssRNA	Mumps RT-PCR	24.65	Orthorubulavirus parotidis	Orthorubulavirus parotitidis		Accept	Correct
49	(-)ssRNA	Mumps RT-PCR	24.12	Orthorubulavirus parotidis	Orthorubulavirus parotitidis		Accept	Correct
50	(-)ssRNA	Mumps RT-PCR	23.29	Orthorubulavirus parotidis	Orthorubulavirus parotitidis		Accept	Correct
51	(-)ssRNA	Measles RT-PCR	16.42	Morbillivirus hominis	Morbillivirus hominis		Accept	Correct
52	(-)ssRNA	Measles RT-PCR	24.59	Morbillivirus hominis	Morbillivirus hominis		Accept	Correct
53	(-)ssRNA	Measles RT-PCR	19.01	Morbillivirus hominis	Morbillivirus hominis		Accept	Correct
54	(-)ssRNA	Measles RT-PCR	25.33	Morbillivirus hominis	Morbillivirus hominis		Accept	Correct
55	(-)ssRNA	Measles RT-PCR	21.75	Morbillivirus hominis	Morbillivirus hominis		Accept	Correct
56	(-)ssRNA	Measles RT-PCR	21.65	Morbillivirus hominis	Morbillivirus hominis		Accept	Correct
57	(-)ssRNA	Measles RT-PCR	24.7	Morbillivirus hominis	Morbillivirus hominis		Accept	Correct
58	(-)ssRNA	Measles RT-PCR	20.01	Morbillivirus hominis	Morbillivirus hominis		Accept	Correct
59	(-)ssRNA	Measles RT-PCR	19.45	Morbillivirus hominis	Morbillivirus hominis		Accept	Correct
60	(-)ssRNA	Measles RT-PCR	22.56	Morbillivirus hominis	Morbillivirus hominis		Accept	Correct
61	(+)ssRNA	Copies/μl	3520.3	Orthoflavivirus nilense	Orthoflavivirus nilense		Accept	Correct
62	(+)ssRNA	Copies/μl	10164	Orthoflavivirus nilense	Orthoflavivirus nilense		Accept	Correct
63	(+)ssRNA	Copies/μl	11505.6	Orthoflavivirus nilense	Orthoflavivirus nilense		Accept	Correct
64	(+)ssRNA	Copies/μl	11506	Orthoflavivirus nilense	Orthoflavivirus nilense		Accept	Correct
65	(+)ssRNA	Copies/μl	11011.6	Orthoflavivirus nilense	Orthoflavivirus nilense		Accept	Correct
66	(+)ssRNA	Copies/μl	12349.5	Orthoflavivirus nilense	Orthoflavivirus nilense		Accept	Correct
67	(+)ssRNA	Copies/μl	5911.3	Orthoflavivirus nilense	Orthoflavivirus nilense		Accept	Correct
68	(+)ssRNA	Ct	22.2	Orthoflavivirus nilense	Orthoflavivirus nilense		Accept	Correct
69	(+)ssRNA	Ct	18	Orthoflavivirus nilense	Orthoflavivirus nilense		Accept	Correct
70	(+)ssRNA	Ct	18.3	Orthoflavivirus nilense	QC Failed		Reject	N / A
76	Segmented (-)ssRNA	Flu_A_H3 Crt 1	6.98 / 6.93	Alphainfluenzavirus influenzae	Segment 1	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 2	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 3	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 4	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 5	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 6	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 7	Alphainfluenzavirus influenzae	Accept	Correct
					Segment 8	Alphainfluenzavirus influenzae	Accept	Correct
76	(-)ssRNA	RSVA Crt 1	25.26 / 25.46	Orthopneumovirus hominis; Orthopneumovirus bovis	No Assembly		Reject	N / A
76	(+)ssRNA	RV_1of2 Crt 1	24.3 / 24.49	Rhinovirus A; Rhinovirus C; Rhinovirus B; Human rhinovirus sp.; Human enterovirus sp. ex; Antwerp rhinovirus 98/99; Human rhinovirus AMS323; Human rhinovirus C192483; Human rhinovirus C210920; Human rhinovirus C326876; Human rhinovirus C326878; Human rhinovirus N53728; Human rhinovirus N53729; Human rhinovirus W9769; Human rhinovirus W9783	No Assembly		Reject	N / A
77	(+)ssRNA	SARS-CoV-2_S Crt 1	25.99 / 27.09	Severe acute respiratory syndrome coronavirus 2	No Assembly		Reject	N / A
77	(+)ssRNA	CoV_229E Crt 1	7.07 / 7.24	Human coronavirus 229E	Human coronavirus 229E		Accept	Correct

78	(+)ssRNA	RV_1of2 Crt 1	18.64 / 18.61	Rhinovirus A; Rhinovirus C; Rhinovirus B; Human rhinovirus sp.; Human enterovirus sp. ex; Antwerp rhinovirus 98/99; Human rhinovirus AMS323; Human rhinovirus C192483; Human rhinovirus C210920; Human rhinovirus C326876; Human rhinovirus C326878; Human rhinovirus N53728; Human rhinovirus N53729; Human rhinovirus W9769; Human rhinovirus W9783	No Assembly	Reject	N / A
78	dsDNA	Adv_1of2 Crt 1	4.78 / 4.83	Human mastadenovirus B; Human mastadenovirus C; Human mastadenovirus D	[Human mastadenovirus C; Human adenovirus 108; Human adenovirus sp.]	Accept	Correct
79	(+)ssRNA	RV_1of2 Crt 1	18.23 / 18.56	Rhinovirus A; Rhinovirus C; Rhinovirus B; Human rhinovirus sp.; Human enterovirus sp. ex; Antwerp rhinovirus 98/99; Human rhinovirus AMS323; Human rhinovirus C192483; Human rhinovirus C210920; Human rhinovirus C326876; Human rhinovirus C326878; Human rhinovirus N53728; Human rhinovirus N53729; Human rhinovirus W9769; Human rhinovirus W9783	No Assembly	Reject	N / A
79	(-)ssRNA	RSVA Crt 1	6.79 / 6.79	Orthopneumovirus hominis; Orthopneumovirus bovis	Orthopneumovirus hominis	Accept	Correct
80	(-)ssRNA	hPIV3_hPIV4 Crt 1	4.75 / 6.8	Respirovirus pneumoniae; Orthorubulavirus hominis	Respirovirus pneumoniae	Accept	Correct
80	(+)ssRNA	RV_1of2 Crt 1	25.1 / 25.33	Rhinovirus A; Rhinovirus C; Rhinovirus B; Human rhinovirus sp.; Human enterovirus sp. ex; Antwerp rhinovirus 98/99; Human rhinovirus AMS323; Human rhinovirus C192483; Human rhinovirus C210920; Human rhinovirus C326876; Human rhinovirus C326878; Human rhinovirus N53728; Human rhinovirus N53729; Human rhinovirus W9769; Human rhinovirus W9783	No Assembly	Reject	N / A
81	(-)ssRNA	RSVB Crt 1	14.83 / 14.74	Orthopneumovirus hominis; Orthopneumovirus bovis	No Assembly	Reject	N / A
81	(-)ssRNA	hPIV3_hPIV4 Crt 1	5.8 / 5.74	Respirovirus pneumoniae; Orthorubulavirus hominis	Respirovirus pneumoniae	Accept	Correct
82	(+)ssRNA	CoV_OC43 Crt 1	20.77 / 20.78	Betacoronavirus 1	No Assembly	Reject	N / A
82	(+)ssRNA	SARS-CoV-2_Orf1b Crt 1	5.11 / 5.17	Severe acute respiratory syndrome coronavirus 2	Severe acute respiratory syndrome coronavirus 2	Accept	Correct
83	Segmented (-)ssRNA	Flu_A_pan Crt 1	24.31 / 24.04	Alphainfluenzavirus influenzae	No Assembly	Reject	N / A
83	(+)ssRNA	SARS-CoV-2_Orf1b Crt 1	5.36 / 5.41	Severe acute respiratory syndrome coronavirus 2	Severe acute respiratory syndrome coronavirus 2	Accept	Correct
84	(+)ssRNA	RV_1of2 Crt 1	23.75 / 23.96	Rhinovirus A; Rhinovirus C; Rhinovirus B; Human rhinovirus sp.; Human enterovirus sp. ex; Antwerp rhinovirus 98/99; Human rhinovirus AMS323; Human rhinovirus C192483; Human rhinovirus C210920; Human rhinovirus C326876; Human rhinovirus C326878; Human rhinovirus N53728; Human rhinovirus N53729; Human rhinovirus W9769; Human rhinovirus W9783	No Assembly	Reject	N / A

				rhinovirus W9769; Human rhinovirus W9783			
84	(+)ssRNA	SARS-CoV-2_Orf1b Crt 1	3.96 / 3.95	Severe acute respiratory syndrome coronavirus 2	Severe acute respiratory syndrome coronavirus 2		Accept Correct
85	(-)ssRNA	hPIV1_hPIV2 Crt 1	9.68 / 9.36	Respirovirus laryngotracheitidis; Orthorubulavirus laryngotracheitidis	Orthorubulavirus laryngotracheitidis		Accept Correct
85	(+)ssRNA	RV_1of2 Crt 1	27.03 / 27.74	Rhinovirus A; Rhinovirus C; Rhinovirus B; Human rhinovirus sp.; Human enterovirus sp. ex; Antwerp rhinovirus 98/99; Human rhinovirus AMS323; Human rhinovirus C192483; Human rhinovirus C210920; Human rhinovirus C326876; Human rhinovirus C326878; Human rhinovirus N53728; Human rhinovirus N53729; Human rhinovirus W9769; Human rhinovirus W9783	No Assembly		Reject N / A
86	(+)ssRNA	CoV_HKU1 Crt 1	8.66 / 8.64	Human coronavirus HKU1	Human coronavirus HKU1		Accept Correct
86	(+)ssRNA	RV_1of2 Crt 1	18.88 / 18.83	Rhinovirus A; Rhinovirus C; Rhinovirus B; Human rhinovirus sp.; Human enterovirus sp. ex; Antwerp rhinovirus 98/99; Human rhinovirus AMS323; Human rhinovirus C192483; Human rhinovirus C210920; Human rhinovirus C326876; Human rhinovirus C326878; Human rhinovirus N53728; Human rhinovirus N53729; Human rhinovirus W9769; Human rhinovirus W9783	No Assembly		Reject N / A
87	Segmented (-)ssRNA	Flu_A_H3 Crt 1	8.37 / 8.46	Alphainfluenzavirus influenzae	Segment 1	Alphainfluenzavirus influenzae	Accept Correct
					Segment 2	Alphainfluenzavirus influenzae	Accept Correct
					Segment 3	Alphainfluenzavirus influenzae	Accept Correct
					Segment 4	Alphainfluenzavirus influenzae	Accept Correct
					Segment 5	Alphainfluenzavirus influenzae	Accept Correct
					Segment 6	Alphainfluenzavirus influenzae	Accept Correct
					Segment 7	Alphainfluenzavirus influenzae	Accept Correct
					Segment 8	Alphainfluenzavirus influenzae	Accept Correct
87	(+)ssRNA	RV_1of2 Crt 1	24.06 / 24.45	Rhinovirus A; Rhinovirus C; Rhinovirus B; Human rhinovirus sp.; Human enterovirus sp. ex; Antwerp rhinovirus 98/99; Human rhinovirus AMS323; Human rhinovirus C192483; Human rhinovirus C210920; Human rhinovirus C326876; Human rhinovirus C326878; Human rhinovirus N53728; Human rhinovirus N53729; Human rhinovirus W9769; Human rhinovirus W9783	No Assembly		Reject N / A
88	(-)ssRNA	RSVB Crt 1	6.27 / 6.26	Orthopneumovirus hominis; Orthopneumovirus bovis	[Orthopneumovirus hominis; Orthopneumovirus bovis]		Accept Correct
88	(+)ssRNA	RV_1of2 Crt 1	20.64 / 20.32	Rhinovirus A; Rhinovirus C; Rhinovirus B; Human rhinovirus sp.; Human enterovirus sp. ex; Antwerp rhinovirus 98/99; Human rhinovirus AMS323; Human rhinovirus C192483; Human rhinovirus C210920; Human rhinovirus C326876; Human rhinovirus C326878; Human rhinovirus N53728; Human rhinovirus N53729; Human	No Assembly		Reject N / A

				rhinovirus W9769; Human rhinovirus W9783			
89	(-)ssRNA	hPIV3_hPIV4 Crt 1	4.93 / 5.06	Respirovirus pneumoniae; Orthorubulavirus hominis	Respirovirus pneumoniae	Accept	Correct
89	(+)ssRNA	RV_1of2 Crt 1	26.51 / 26.54	Rhinovirus A; Rhinovirus C; Rhinovirus B; Human rhinovirus sp.; Human enterovirus sp. ex; Antwerp rhinovirus 98/99; Human rhinovirus AMS323; Human rhinovirus C192483; Human rhinovirus C210920; Human rhinovirus C326876; Human rhinovirus C326878; Human rhinovirus N53728; Human rhinovirus N53729; Human rhinovirus W9769; Human rhinovirus W9783	No Assembly	Reject	N / A
90	(-)ssRNA	hPIV3_hPIV4 Crt 1	6.72 / 6.79	Respirovirus pneumoniae; Orthorubulavirus hominis	Respirovirus pneumoniae	Accept	Correct
90	(-)ssRNA	RSVA Crt 1	25.27 / 25.32	Orthopneumovirus hominis; Orthopneumovirus bovis	No Assembly	Reject	N / A
Taxonomy Classification Accuracy						168 / 185 (91%)	168 / 168 (100%)

*CRT values are determined using the Thermo Fisher Open Array protocol, they are equivalent to qRT-PCR Ct values minus 10 cycles.

**Acceptable taxonomic results are separated by a semi-colon.

Genome Assembly

Next, to validate the assay can reliably generate viral genome assemblies that are identical to the reference sequence, we compared 9 clinical samples and 17 contrived samples to their ground truth reference sequences from NCBI or GISAID, generated using validated methods. The 9 clinical samples include 8 SARS-CoV-2 cases and 1 Respiratory Syncytial Virus case, where their reference sequences were generated using a validated method by Lea Starita lab at University of Washington. Acknowledging the limitation in the diversity of clinical samples, we also included 17 contrived samples obtained from Twist Biosciences and BEI Resources - each represents a unique virus species (**Table 2**). The accuracy of genome assembly by VAPER was evaluated as the percentage of assemblies sharing at least 99.5% nucleotide identity with the corresponding ground truth sequences listed in **Table 2**. Only samples that passed the acceptance/rejection criteria were considered. Nucleotide positions in the test sample reported as unknown, as represented by nucleotide code N, will not be considered.

In total, 26 samples were compared, including 3 subtypes of Influenza viruses (Influenza A virus H1N1, H3N2 and Influenza B virus). For segmented influenza virus genomes, each genome segment was assembled and compared to their corresponding reference sequence individually. In addition, Sample #74, #75, #82, #83 and #84 contain low-abundance co-infecting viruses, indicated by qRT-PCR results. However, these viruses are excluded from the comparison as their Crt values failed to meet sample acceptance criteria and lacked existing reference sequences. During this part of the validation, we identified that Sample #99, Twist synthetic control for Bocavirus, only contains partial genome and excluded this sample from comparison [JJ1] (please see attached email communications from Illumina and Twist). In summary, out of the 46 genome assemblies compared, 44 assemblies passed the 99.5% nucleotide identity threshold resulting in 95.7% accuracy.

Reasons for assembly accuracy failure were investigated further. Sample #91 failed to meet the assembly accuracy criteria due to 27 disagreements between the test assembly and the truth assembly. 24 of these disagreements were due to a large insertion at the 3' end of the sequence (position 1700). As shown in **Figure 5**, this false insertion resulted from the spurious alignment of a relatively small subset of off-target reads. While some spurious alignments are expected when mapping reads to a reference genome, these errors are not normally problematic due to their relative infrequency compared to correct alignments. However, the tremendous read depth observed

in Sample #91 (i.e., ~ 20,000X average coverage) resulted in an increased likelihood of spurious alignments, which could not be distinguished from correct, on-target alignments. Fortunately, this issue is easily resolved by re-running the assembly using a more restrictive maximum read limit (default: “--max_reads 2000000”). As described in the SOP (MediaLab Document #69790.5317) for this assay, assemblies with excessive read depth (> 200X) can be re-assembled using a read limit that will ensure the assembly does not exceed 200X coverage. Applying this method to the Sample #91 assembly resulted in the elimination of the spurious section of the genome, resulting in ~99.9% agreement with the truth assembly within comparable regions (100% of the genome).

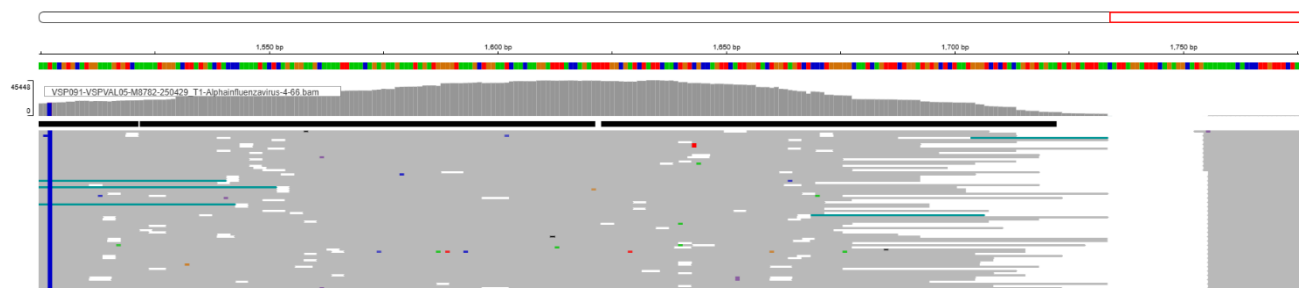


Figure 5. Alignment of the Sample #91 paired-end reads to the Alphainfluenzavirus-4-66 reference assembly. Spurious alignments resulting from excessive sequencing depth can be observed around nucleotide 1750 in the reference assembly.

Sample #104 failed to meet the assembly accuracy criteria due to 2534 disagreements with the truth assembly. Roughly 77% of these differences occurred within inverted terminal repeat (ITR) regions located between positions ~1-9400 and 190000-200000 of the Monkeypox virus (MPXV) genome (Desingu PA et al., 2023). While these differences could be errors in the VAPER assembly, it is also important to consider that the truth assembly used for Sample #104 (NCBI accession AY603973.1) was created in 2005 (> 20 years ago), using early genome sequencing and assembly methods (Chen et al., 2005). [JJ2] As such, it is also likely that the truth assembly contains errors within these repeat regions, making it very difficult to draw an accurate conclusion. Future investigations with long read sequencing, such as Oxford Nanopore, are recommended to resolve these discrepancies. Until then, Monkeypox virus genomes will be masked between positions 1-9400 and 190000-200000 prior to reporting SOP (MediaLab Document #69790.5317). Applying this masking to Sample #104 resulted in ~99.7% agreement with the comparable regions within the truth assembly (> 90% of the genome).

Table 2: Genome assembly accuracy reporting table

Sample #	Virus	Virus Subtype	Viral Genome Form	Reference Sequence Source and Accession	Analysis QC	Genome Assembly Identity (%)	Interpretation (>99.5%)
19	Coronavirus	SARS-CoV-2	(+)ssRNA	NCBI: ON808942	Accept	99.97	Correct
71	Coronavirus	SARS-COV-2	(+)ssRNA	NCBI: ON207300	Accept	99.97	Correct
72	Coronavirus	SARS-COV-2	(+)ssRNA	NCBI: OP755407	Accept	99.61	Correct
73	Respiratory syncytial virus	B	(-)ssRNA	NCBI: PP748817	Accept	99.8	Correct
74	Coronavirus	SARS-COV-2	(+)ssRNA	NCBI: ON808850	Accept	99.97	Correct
	Rhinovirus		(+)ssRNA	N/A*	Reject	No Assembly	N / A
75	Coronavirus	SARS-COV-2	(+)ssRNA	NCBI: ON207261	Accept	99.72	Correct
	Rhinovirus		(+)ssRNA	N/A*	Reject	No Assembly	N / A
82	Coronavirus	CoV-OC43	(+)ssRNA	N/A*	Reject	No Assembly	N / A
	Coronavirus	SARS-CoV-2	(+)ssRNA	NCBI: OP860708	Accept	99.95	Correct
83	Influenza	Unknown	Segmented (-)ssRNA	N/A*	Reject	No Assembly	N / A
	Coronavirus	SARS-CoV-2	(+)ssRNA	NCBI: OQ230605	Accept	99.97	Correct
84	Rhinovirus		(+)ssRNA	N/A*	Reject	No Assembly	N / A
	Coronavirus	SARS-CoV-2	(+)ssRNA	NCBI: NC207350	Accept	99.96	Correct
91	Influenza	H1N1	Segmented (-)ssRNA	NCBI: NC_026438	Accept	100	Correct

				NCBI: NC_026435	Accept	99.91	Correct
				NCBI: NC_026437	Accept	99.91	Correct
				NCBI: NC_026433	Accept	98.41	Incorrect
				NCBI: NC_026436	Accept	99.93	Correct
				NCBI: NC_026434	Accept	100	Correct
				NCBI: NC_026431	Accept	100	Correct
92	Influenza	H3N2	Segmented (-)ssRNA	NCBI: NC_026432	Accept	100	Correct
				NCBI: NC_007373	Accept	100	Correct
				NCBI: NC_007372	Accept	100	Correct
				NCBI: NC_007371	Accept	100	Correct
				NCBI: NC_007366	Accept	100	Correct
				NCBI: NC_007369	Accept	100	Correct
				NCBI: NC_007368	Accept	99.93	Correct
				NCBI: NC_007367	Accept	100	Correct
93	Influenza	IBV	Segmented (-)ssRNA	NCBI: NC_007370	Accept	99.76	Correct
				NCBI: NC_002204	Accept	100	Correct
				NCBI: NC_002205	Accept	99.91	Correct
				NCBI: NC_002206	Accept	99.91	Correct
				NCBI: NC_002207	Accept	100	Correct
				NCBI: NC_002208	Accept	99.89	Correct
				NCBI: NC_002209	Accept	100	Correct
				NCBI: NC_002210	Accept	99.83	Correct
94	Enterovirus	EV68	(+)ssRNA	NCBI: NC_002211	Accept	100	Correct
				NCBI: NC_038308.1	Accept	100	Correct
95	Mumps virus		(-)ssRNA	NCBI: NC_002200.1	Accept	99.99	Correct
96	Measles virus		(-)ssRNA	NCBI: NC_001498.1	Accept	99.99	Correct
97	Parainfluenza virus	HPIV1	(-)ssRNA	NCBI: NC_003461.1	Accept	100	Correct
98	Parainfluenza virus	HPIV4	(-)ssRNA	NCBI: NC_021928.1	Accept	100	Correct
99	Bocavirus	HBoV	ssDNA	NCBI: MG953830.1	Reject	Failed QC**	N / A
100	Coronavirus	CoV-229E	(+)ssRNA	NCBI: NC_002645.1	Accept	99.86	Correct
101	Coronavirus	CoV-NL63	(+)ssRNA	NCBI: NC_005831.2	Accept	100	Correct
102	Coronavirus	CoV-OC43	(+)ssRNA	NCBI: NC_006213.1	Accept	99.97	Correct
103	Coronavirus	SARS-CoV-2 BA.4	(+)ssRNA	GISAID: EPI_ISL_12454576	Accept	99.99	Correct
104	Monkeypox Virus	WRAIR 7-61	dsDNA	NCBI: AY603973.1	Accept	98.71	Incorrect
105	Hepatitis A Virus	HM175/18f	(+)ssRNA	NCBI: KP879217.1	Accept	99.95	Correct
106	Respiratory syncytial virus	A1998/12-21	(-)ssRNA	NCBI: JX069802	Accept	99.97	Correct
108	Metapneumovirus	TN/91-316	(-)ssRNA	NCBI: KC403971	Accept	99.79	Correct
Genome Assembly Accuracy					46/52 (88.46%)	44/46 (95.65%)	

* Additional viruses detected in the samples via qRT-PCR, but the sample Crt failed to meet acceptance criteria, and the reference sequence is not available. The genome assemblies of these viruses will not be compared.

** This synthetic control sample is missing a fraction of the genome. It is thus rejected from validation.

Together, using the combined clinical, environmental and contrived samples, we sought to overcome the challenge of limited reference sequence, while evaluating all aspects of the Illumina Viral Surveillance kit and the VAPER bioinformatic analysis pipeline. Importantly, this validation included viruses of diverse genome forms (single-stranded DNA [ssDNA], double-stranded DNA [dsDNA], positive sense single stranded RNA [(+)ssRNA], and negative sense single stranded RNA [(-)ssRNA], with both segmented and non-segmented genomes), and diverse molecular structures, including enveloped, non-enveloped and/or encapsulated viruses. The collection of the viruses is biochemically and structurally representative of the remaining 200 different viruses included in the hybrid-capture probe panel (Quek et al. 2024). Given the broad compatibility of the total nucleic acid extraction method and the library preparation techniques, we reason that this assay has the potential to provide genome sequencing data for a wide array of viruses of public health concerns.

REPRODUCIBILITY: Can you obtain the same result time after time? Confirming this parameter by repeatedly testing the same samples on different days using different computers to interface with the bioinformatic pipeline and getting the same or comparable results (reproducible), regardless of which member of the laboratory's testing personnel performs the test (operator variance).

Fully automated test systems that are not operator dependent, operator variance should not affect the test's precision and may not need to be evaluated by more than one person. Acceptable precision is considered calculated reproducibility $\geq 95\%$.

$$\text{Reproducibility} = [\text{No. of repeated results in agreement} / \text{Total no. results}] \times 100$$

$$\text{Taxonomy Classification Reproducibility} = 100 \times \frac{21}{21} = 100\%$$

$$\text{Genome Assembly Reproducibility} = 100 \times \frac{21}{21} = 100\%$$

For reproducibility evaluation, a subset of 7 clinical and environmental samples, including one sample representing each unique viral infection (Adenovirus, Coronavirus, Influenza virus, Parainfluenza virus, Mumps virus, Measles virus and West Nile virus) were sequenced along with 1 negative control (water) and 1 positive control (synthetic viral genome). This set of samples was independently extracted and sequenced by three technicians using the Illumina Viral Surveillance kit on three separate days. These sequencing libraries were sequenced on three separate MiSeq instruments and analyzed via the VAPER bioinformatic pipeline. Reproducibility was determined as the number of repeated results in agreement over the total number of comparisons between repeated results. A pair of repeated taxonomy results will be considered in agreement if the reported taxonomy is identical. A pair of repeated assembly results will be considered in agreement if they share at least 99.5% nucleotide identity in comparable regions (i.e., those not reported as an N). Together, all three replicates reproduced the nearly identical sequencing results. Through comparisons of the 7 samples times among 3 pair-wise comparison, the validation showed 100% reproducibility in taxonomy classification and genome assembly (Table 3).

Table 3: Reproducibility reporting table

Sample #	Virus	Viral Genome Form	Observed Taxonomic Result (Run #1 vs #2)		Genome Assembly Identity (% Run #1 vs #2)	Observed Taxonomic Result (Run #2 vs #3)		Genome Assembly Identity (% Run #2 vs #3)	Observed Taxonomic Result (Run #1 vs #3)		Genome Assembly Identity (% Run #1 vs #3)	Taxonomic results in agreement	Genome assembly results in agreement (>99.5% Identity)
			Run 1	Run 2		Run 2	Run 3		Run 1	Run 3			
1	Adenovirus	dsDNA	[Human mastadenovirus B; Human adenovirus sp.]	[Human mastadenovirus B; Human adenovirus sp.]	100.00	[Human mastadenovirus B; Human adenovirus sp.]	[Human mastadenovirus B; Human adenovirus sp.]	100.00	[Human mastadenovirus B; Human adenovirus sp.]	[Human mastadenovirus B; Human adenovirus sp.]	100.00	3/3 (100%)	3/3 (100%)
11	Coronavirus 229E	(+)ssRNA	Human coronavirus 229E	Human coronavirus 229E	100.00	Human coronavirus 229E	Human coronavirus 229E	100.00	Human coronavirus 229E	Human coronavirus 229E	100.00	3/3 (100%)	3/3 (100%)
21	Influenza A H1	Segmented (-)ssRNA	Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	100.00	Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	100.00	Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	100.00	3/3 (100%)	3/3 (100%)
			Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	100.00	Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	100.00	Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	100.00	3/3 (100%)	3/3 (100%)
			Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	100.00	Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	100.00	Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	100.00	3/3 (100%)	3/3 (100%)
			Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	100.00	Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	100.00	Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	100.00	3/3 (100%)	3/3 (100%)
			Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	99.94	Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	99.94	Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	99.94	3/3 (100%)	3/3 (100%)
			Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	100.00	Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	100.00	Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	100.00	3/3 (100%)	3/3 (100%)

			Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	100.00	Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	100.00	Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	100.00	3/3 (100%)	3/3 (100%)
			Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	100.00	Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	100.00	Alphainfluenzavirus influenzae	Alphainfluenzavirus influenzae	100.00	3/3 (100%)	3/3 (100%)
36	Parainfluenza 3/4	(-)ssRNA	Respirovirus pneumoniae	Respirovirus pneumoniae	99.98	Respirovirus pneumoniae	Respirovirus pneumoniae	99.98	Respirovirus pneumoniae	Respirovirus pneumoniae	99.98	3/3 (100%)	3/3 (100%)
41	Mumps virus	(-)ssRNA	Orthorubulavirus parotitidis	Orthorubulavirus parotitidis	100.00	Orthorubulavirus parotitidis	Orthorubulavirus parotitidis	100.00	Orthorubulavirus parotitidis	Orthorubulavirus parotitidis	100.00	3/3 (100%)	3/3 (100%)
51	Measles virus	(-)ssRNA	Morbillivirus hominis	Morbillivirus hominis	100.00	Morbillivirus hominis	Morbillivirus hominis	100.00	Morbillivirus hominis	Morbillivirus hominis	100.00	3/3 (100%)	3/3 (100%)
66	West Nile Virus	(+)ssRNA	Orthoflavivirus nilense	Orthoflavivirus nilense	100.00	Orthoflavivirus nilense	Orthoflavivirus nilense	100.00	Orthoflavivirus nilense	Orthoflavivirus nilense	100.00	3/3 (100%)	3/3 (100%)
Total Number of Replicates in Agreement (%)												21/21 (100%)	21/21 (100%)

ANALYTICAL SENSITIVITY: (Qualitative Test only) Analytical Sensitivity measures the number of quantifiable samples at specified concentration levels near the expected detection limit.

$$\text{Sensitivity} = [\text{TP}/(\text{TP}+\text{FN})] \times 100$$

$$\text{Specificity} = [\text{TN}/(\text{TN}+\text{FP})] \times 100.$$

Calculations for these should come from your accuracy sample pool numbers (TP, FP, TN, FN). Both sensitivity and specificity should be at least 95% compared to the reference method.

This assay is not intended for diagnostic purposes, so the analytical sensitivity is NOT APPLICABLE to this assay. All samples analyzed by this assay are first tested by Virology or Environmental Laboratory Science divisions. The presence and abundance of viral genome is determined previously using ddPCR, qRT-PCR or other approved diagnostic assays. This proposed assay is meant to produce high quality sequencing data of the viral genomes to identify genetic variants. The resulting categorization of the virus taxonomy produced by the bioinformatic pipeline is only used by epidemiologists at their discretion.

OPTIONAL ANALYSIS METRICS

If the analysis is on samples that represent a population or attempt to detect a target within a pool of potential targets, the following metrics must be addressed in the validation.

LOWER LIMIT OF DETECTION (LOD):

LOD may be influenced by genome size and is required for each variant and microbial target type. For antiviral drug resistance, LOD establishment must consider the virus load and variant allele fraction.

All clinical and environmental samples used for validation and future studies are first evaluated using ddPCR, qRT-PCR and other quantitative methods. These methods are previously validated independently, and the limits of detection are previously determined. The Advanced Molecular Detection Lab only processes samples that meet the LOD and the Input sample rejection criteria. Furthermore, we established strict post-assembly rejection criteria that are more stringent than assay LOD, so we expect a small subset of the samples to generate suboptimal genome assembly. We will re-sequence or eliminate samples that failed these additional criteria to ensure that high quality sequencing data and genome assembly are obtained.

LITERATURE REFERENCES/PACKAGE INSERTS: List **below** any technical and/or literature used in the verification planning process.

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- Illumina Viral Surveillance Panel v2 Kit Data Sheet: <https://www.illumina.com/content/dam/illumina/gcs/assembled-assets/marketing-literature/viral-surveillance-panel-v2-data-sheet-m-gl-02882/viral-surveillance-panel-v2-data-sheet-m-gl-02882.pdf>. Accessed on December 31, 2024.
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NEED FOR CLIENT EDUCATION: Please describe **below** how this will be provided **if needed**.

- Are there any clinical performance claims? **None**

Appendix A: Samples

Sample #	Source	ID	Virus	Virus Subtype	Viral Genome Form
1	Starita	cb800476	Adenovirus		dsDNA
2	Starita	3e683cca	Adenovirus		dsDNA
3	Starita	a3fdb50c	Adenovirus		dsDNA
4	Starita	c3d00906	Adenovirus		dsDNA
5	Starita	0747406c	Adenovirus		dsDNA
6	Starita	218fca8a	Adenovirus		dsDNA
7	Starita	5c31fc37	Adenovirus		dsDNA
8	Starita	d2196e74	Adenovirus		dsDNA
9	Starita	4975f05e	Adenovirus		dsDNA
10	Starita	ae2c2005	Adenovirus		dsDNA
11	Starita	80615f50	Coronavirus	CoV_229E	(+)ssRNA
12	Starita	0fa35ea0	Coronavirus	CoV_229E	(+)ssRNA
13	Starita	29fd5427	Coronavirus	CoV_HKU1	(+)ssRNA
14	Starita	1e7e621e	Coronavirus	CoV_HKU1	(+)ssRNA
15	Starita	c708be10	Coronavirus	CoV_NL63	(+)ssRNA
16	Starita	a9a6b7ec	Coronavirus	CoV_NL63	(+)ssRNA
17	Starita	c4f34162	Coronavirus	CoV_OC43	(+)ssRNA
18	Starita	fdb2db15	Coronavirus	CoV_OC43	(+)ssRNA
19	Starita	e4927e5b	Coronavirus	SARS-CoV-2	(+)ssRNA
20	Starita	fd09b5c3	Coronavirus	SARS-CoV-2	(+)ssRNA
21	Starita	52d7b35c	Influenza	A-H1	Segmented (-)ssRNA
22	Starita	e5277ba4	Influenza	A-H1	Segmented (-)ssRNA
23	Starita	a3f48369	Influenza	A-H1	Segmented (-)ssRNA
24	Starita	0231ba0a	Influenza	A-H3	Segmented (-)ssRNA
25	Starita	74314522	Influenza	A-H3	Segmented (-)ssRNA
26	Starita	d0b7dc79	Influenza	A-H3	Segmented (-)ssRNA
27	Starita	c3f77ba6	Influenza	B	Segmented (-)ssRNA
28	Starita	6a443e12	Influenza	B	Segmented (-)ssRNA
29	Starita	6d2f401e	Influenza	B	Segmented (-)ssRNA
30	Starita	cd7b3a01	Influenza	B	Segmented (-)ssRNA
31	Starita	e023b13e	Parainfluenza	1 or 2	(-)ssRNA
32	Starita	0bad6d5b	Parainfluenza	1 or 2	(-)ssRNA
33	Starita	054077e9	Parainfluenza	1 or 2	(-)ssRNA
34	Starita	57489e5f	Parainfluenza	1 or 2	(-)ssRNA
35	Starita	a84d1edd	Parainfluenza	1 or 2	(-)ssRNA
36	Starita	3bcdbf17	Parainfluenza	3 or 4	(-)ssRNA
37	Starita	62e3342c	Parainfluenza	3 or 4	(-)ssRNA
38	Starita	e5a0f77e	Parainfluenza	3 or 4	(-)ssRNA
39	Starita	caa6389c	Parainfluenza	3 or 4	(-)ssRNA
40	Starita	1d459c9e	Parainfluenza	3 or 4	(-)ssRNA
41	Virology	WA0268049	Mumps virus		(-)ssRNA
42	Virology	WA0268501	Mumps virus		(-)ssRNA
43	Virology	WA0267407	Mumps virus		(-)ssRNA
44	Virology	WA0267408	Mumps virus		(-)ssRNA
45	Virology	WA0267551	Mumps virus		(-)ssRNA
46	Virology	WA0268742	Mumps virus		(-)ssRNA
47	Virology	WA0268864	Mumps virus		(-)ssRNA
48	Virology	WA0268862	Mumps virus		(-)ssRNA

49	Virology	WA0269142	Mumps virus		(-)ssRNA
50	Virology	WA0269651	Mumps virus		(-)ssRNA
51	Virology	WA0232146	Measles virus		(-)ssRNA
52	Virology	WA0232673	Measles virus		(-)ssRNA
53	Virology	WA0232672	Measles virus		(-)ssRNA
54	Virology	WA0232213	Measles virus		(-)ssRNA
55	Virology	WA0222841	Measles virus		(-)ssRNA
56	Virology	WA0221758	Measles virus		(-)ssRNA
57	Virology	WA0221951	Measles virus		(-)ssRNA
58	Virology	WA0221760	Measles virus		(-)ssRNA
59	Virology	WA0221759	Measles virus		(-)ssRNA
60	Virology	WA0983473	Measles virus		(-)ssRNA
61	ELS	WA1015120	West Nile Virus		(+)ssRNA
62	ELS	WA1012615	West Nile Virus		(+)ssRNA
63	ELS	WA1010119	West Nile Virus		(+)ssRNA
64	ELS	WA1007744	West Nile Virus		(+)ssRNA
65	ELS	WA1007633	West Nile Virus		(+)ssRNA
66	ELS	WA1007064	West Nile Virus		(+)ssRNA
67	ELS	WA1004848	West Nile Virus		(+)ssRNA
68	ELS	WA0942103	West Nile Virus		(+)ssRNA
69	ELS	WA0949891	West Nile Virus		(+)ssRNA
70	ELS	WA0949896	West Nile Virus		(+)ssRNA
71	Starita	2545a2ef	Coronavirus	SARS-COV-2	(+)ssRNA
72	Starita	06075d0f	Coronavirus	SARS-COV-2	(+)ssRNA
73	Starita	74bf81c8	Respiratory syncytial virus	B	(-)ssRNA
74	Starita	9d1a3133	Coronavirus	SARS-COV-2	(+)ssRNA
			Rhinovirus		(+)ssRNA
75	Starita	d4ae9817	Coronavirus	SARS-COV-2	(+)ssRNA
			Rhinovirus		(+)ssRNA
76	Starita	2be3d609	Influenza	H3	Segmented (-)ssRNA
			Respiratory syncytial virus	A	(-)ssRNA
			Rhinovirus		(+)ssRNA
77	Starita	c73ad323	Coronavirus	SARS-CoV-2	(+)ssRNA
			Coronavirus	CoV-229E	(+)ssRNA
78	Starita	0747406c	Rhinovirus		(+)ssRNA
			Adenovirus		dsDNA
79	Starita	60146336	Rhinovirus		(+)ssRNA
			Respiratory syncytial virus	A	(-)ssRNA
80	Starita	e1dcae21	Parainfluenza	3 or 4	(-)ssRNA
			Rhinovirus		(+)ssRNA
81	Starita	8a628456	Respiratory syncytial virus	B	(-)ssRNA
			Parainfluenza	3 or 4	(-)ssRNA
82	Starita	9cfda4fd	Coronavirus	CoV-OC43	(+)ssRNA
			Coronavirus	SARS-CoV-2	(+)ssRNA
83	Starita	59b10d28	Influenza	Unknown	Segmented (-)ssRNA
			Coronavirus	SARS-CoV-2	(+)ssRNA
84	Starita	9644b1e1	Rhinovirus		(+)ssRNA
			Coronavirus	SARS-CoV-2	(+)ssRNA
85	Starita	294dd940	Parainfluenza	1 or 2	(-)ssRNA
			Rhinovirus		(+)ssRNA
86	Starita	391c7546	Coronavirus	CoV_HKU1	(+)ssRNA

			Rhinovirus		(+)ssRNA
87	Starita	3426e5bf	Influenza	H3	Segmented (-)ssRNA
			Rhinovirus		(+)ssRNA
88	Starita	e2341e8b	Respiratory syncytial virus	B	(-)ssRNA
			Rhinovirus		(+)ssRNA
89	Starita	c989e8c7	Parainfluenza	3 or 4	(-)ssRNA
			Rhinovirus		(+)ssRNA
90	Starita	07cc76e0	Parainfluenza	3 or 4	(-)ssRNA
			Respiratory syncytial virus	A	(-)ssRNA
91	TWIST Biosciences	SVG-1	Influenza	H1N1	Segmented (-)ssRNA
92	TWIST Biosciences	SVG-2	Influenza	H3N2	Segmented (-)ssRNA
93	TWIST Biosciences	SVG-3	Influenza	IBV	Segmented (-)ssRNA
94	TWIST Biosciences	SVG-4	Enterovirus	EV68	(+)ssRNA
95	TWIST Biosciences	SVG-5	Mumps virus		(-)ssRNA
96	TWIST Biosciences	SVG-6	Measles virus		(-)ssRNA
97	TWIST Biosciences	SVG-7	Parainfluenza virus	HPIV1	(-)ssRNA
98	TWIST Biosciences	SVG-8	Parainfluenza virus	HPIV4	(-)ssRNA
99	TWIST Biosciences	SVG-9	Bocavirus	HBoV	ssDNA
100	TWIST Biosciences	SVG-10	Coronavirus	CoV-229E	(+)ssRNA
101	TWIST Biosciences	SVG-11	Coronavirus	CoV-NL63	(+)ssRNA
102	TWIST Biosciences	SVG-12	Coronavirus	CoV-OC43	(+)ssRNA
103	TWIST Biosciences	SVG-13	Coronavirus	SARS-CoV-2 BA.4	(+)ssRNA
104	BEI Resources	SVG-14	Mpox virus	WRAIR 7-61	dsDNA
105	BEI Resources	SVG-15	Hepatitis A Virus	HM175/18f	(+)ssRNA
106	BEI Resources	SVG-16	Respiratory syncytial virus	A1998/12-21	(-)ssRNA
108	BEI Resources	SVG-18	Human Metapneumovirus	TN/91-316	(-)ssRNA